BE-105 · BTech First Year · UIT, Barkatullah University, Bhopal

Basic Electronics Engineering — Short Notes

Unit-wise complete syllabus coverage · Hinglish notes · Prepared by Ajeet Soni

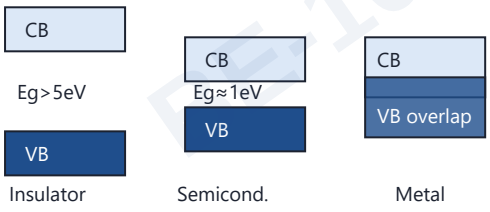
UNIT I Semiconductor Theory & Diodes

1. Energy Band Theory of Crystals

Isolated atom mein electrons ke **discrete energy levels** hote hain. Jab bahut saare atoms paas aakar **crystal** banate hain, to neighbouring atoms ke levels overlap karke **energy bands** ban jaate hain (Pauli exclusion principle ki wajah se ek level split ho jaata hai).

- **Valence Band (VB):** outermost (valence) electrons wala band; 0 K par completely filled.
- **Conduction Band (CB):** free electrons ka band; isme aaye electrons current carry karte hain.
- **Forbidden Energy Gap (E_g):** VB top aur CB bottom ke beech ka gap, jahan koi electron allowed nahi. Unit: eV (1 eV = 1.6×10^{-19} J).

2. Insulators, Semiconductors & Metals



Energy band diagrams

Type	E_g	Example
Insulator	> 5 eV (wide)	Diamond (~5.5 eV), glass
Semiconductor	~1 eV	Si 1.12 eV, Ge 0.72 eV
Metal (conductor)	0 (bands overlap)	Cu, Al, Ag

Semiconductor ka resistivity temperature badhne par **ghatta** hai (negative temperature coefficient), metal ka **badhta** hai.

3. Intrinsic & Extrinsic Semiconductors

Intrinsic (pure) semiconductor

- Pure Si/Ge; room temp par kuch covalent bonds tootte hain → **electron-hole pairs**.
- $n = p = n_i$ (intrinsic concentration). Si: $n_i \approx 1.5\times10^{10}/cm^3$ at 300 K.
- Fermi level E_F lagbhag band gap ke **beech** mein.

Extrinsic (doped) semiconductor

- **N-type:** pentavalent impurity (P, As, Sb) = **donor**. Majority carriers = electrons. E_F CB ke paas.
- **P-type:** trivalent impurity (B, Al, Ga, In) = **acceptor**. Majority = holes. E_F VB ke paas.

FORMULA Mass action law: $n\cdot p = n_i^2$ | Conductivity: $\sigma = q(n\mu_n + p\mu_p)$ | N-type: $n \approx N_D$, $p = n_i^2/N_D$ | Intrinsic: $\sigma_i = q\, n_i(\mu_n+\mu_p)$

4. Theory of P-N Junction

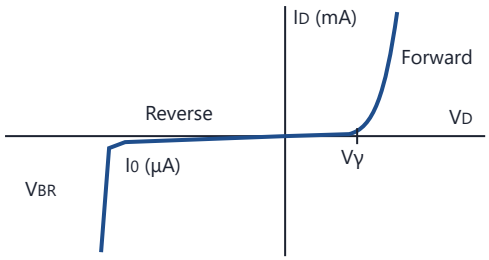
P aur N type ko ek hi crystal mein milane par junction banta hai. Concentration gradient ki wajah se holes P→N aur electrons N→P **diffuse** karte hain aur recombine ho jaate hain. Junction ke paas sirf **immobile ions** bachte hain (P side negative, N side positive) → **depletion region** (space-charge region). Ye ions ek electric field banate hain jo aur diffusion ko rokta hai → **barrier potential V_0** (Si ≈ 0.7 V, Ge ≈ 0.3 V).

FORMULA

$$V_0 = V_T \ln\frac{N_A N_D}{n_i^2},\quad V_T = \frac{kT}{q} = \frac{T}{11600} \approx 26\text{ mV at }300\text{ K}$$

- **Forward bias** (P → +ve): barrier kam hota hai, depletion width ghatti hai, large majority-carrier current (mA).
- **Reverse bias** (P → −ve): barrier badhta hai, depletion width badhti hai, sirf tiny minority-carrier current I_0 (nA–μA) = reverse saturation current.

5. Ideal & Practical Diode; Characteristics & Operation



Practical diode V-I characteristic

- **Ideal diode:** forward mein short circuit (V=0, R=0), reverse mein open circuit (I=0). Perfect switch.
- **Practical diode:** conduction tab hota hai jab V > **cut-in / knee voltage V_γ** (Si 0.7 V, Ge 0.3 V); forward resistance r_f finite hota hai; reverse mein small I_0 flow karta hai; V_{BR} par breakdown.
- Static resistance R = V_D/I_D; dynamic resistance $r_d = \Delta V/\Delta I = \eta V_T/I_D$ (≈ 26 mV/I_D).

DIODE (SHOCKLEY) EQUATION

$$I = I_0\left(e^{V/\eta V_T} - 1\right)$$

η = ideality factor (1 for Ge, ≈2 for Si at low current). Forward large V: $I \approx I_0 e^{V/\eta V_T}$; reverse: $I \approx -I_0$.

6. Temperature & Breakdown Characteristics

Temperature effect

- I_0 har **10 °C** rise par **double**: $I_0(T_2) = I_0(T_1)\cdot 2^{(T_2-T_1)/10}$.
- Same current ke liye V_D ≈ **−2.5 mV/°C** (Si) ghatta hai → forward curve left shift.
- Breakdown voltage temperature ke saath change hota hai (neeche dekho).

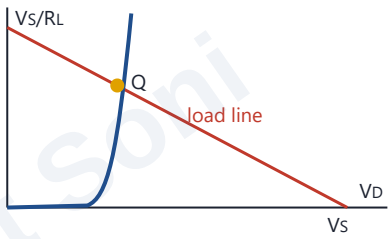
Breakdown mechanisms

Zener breakdown	Avalanche breakdown
Heavily doped, thin depletion layer	Lightly doped, wide depletion layer
Strong field (~10^7 V/m) bonds todta hai (field emission)	Carriers accelerate → impact ionization → chain multiplication
V_Z < ~5 V; **negative** temp. coeff.	V > ~6 V; **positive** temp. coeff.

7. Load Line Analysis

Circuit: source V_S, series resistor R_L, diode. KVL: $V_S = I_D R_L + V_D$. Ye ek straight line hai = **DC load line**.

- $I_D = 0 \rightarrow V_D = V_S$ (x-intercept)
- $V_D = 0 \rightarrow I_D = V_S/R_L$ (y-intercept)
- Load line aur diode curve ka intersection = **Q-point** (operating point: I_{DQ}, V_{DQ}).



Q-point = intersection

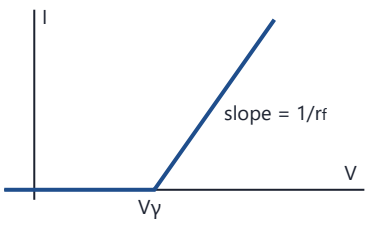
8. Junction Capacitances

Transition / Depletion capacitance C_T	Diffusion / Storage capacitance C_D
Reverse bias mein dominant. Depletion layer = dielectric, P & N regions = plates.	Forward bias mein dominant. Injected minority charge store hota hai.
$C_T = \varepsilon A/W$; $W \propto (V_0+V_R)^n$ → V_R badhe to C_T ghate (pF range)	$C_D = \tau I/(\eta V_T)$; current ke proportional (nF–μF range), τ = mean lifetime

9. Piecewise Linear DC Model of Diode

Exponential curve ko straight-line segments se approximate karte hain:

1. **Ideal model:** V_γ=0, r_f=0 (switch).
2. **Constant voltage drop model:** ON → battery V_γ (0.7 V); OFF → open.
3. **Piecewise linear model:** ON (V ≥ V_γ) → V_γ battery + r_f series; OFF (V < V_γ) → open circuit. $I = (V - V_\gamma)/r_f$.



Piecewise linear approximation

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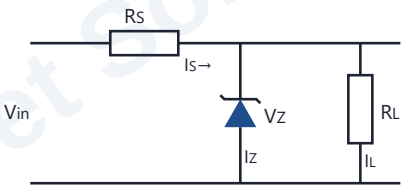
EXAM TIP Numericals mein jab kuch na diya ho to Si ke liye V_γ = 0.7 V, Ge ke liye 0.3 V lo. Example: V_S = 10 V, R = 1 kΩ, Si diode, r_f = 10 Ω → I = (10 − 0.7)/(1000 + 10) = 9.21 mA.

UNIT II Applications of Diode & Wave Shaping Circuits

1. Zener Diode & Design of Zener Regulator

Zener diode heavily doped P-N diode hai jo **reverse breakdown region** mein operate karne ke liye design hota hai. Breakdown mein current kaafi change hone par bhi voltage ≈ V_Z constant rehta hai → **voltage regulator / reference**.

- Parameters: V_Z, $I_{Z(min)}$ (= I_{ZK}, knee), $I_{Z(max)} = P_Z/V_Z$, zener resistance r_Z.



Zener shunt regulator: $I_S = I_Z + I_L$

DESIGN FORMULAS

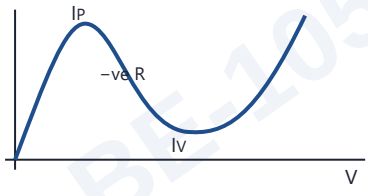
$$R_S = \frac{V_{in} - V_Z}{I_Z + I_L} \;;\; R_{S(max)} = \frac{V_{in(min)} - V_Z}{I_{Z(min)} + I_{L(max)}} \;;\; R_{S(min)} = \frac{V_{in(max)} - V_Z}{I_{Z(max)} + I_{L(min)}}$$

Zener ON rahe iske liye: $V_L = V_{in}R_L/(R_S+R_L) \geq V_Z$. Zener power $P_Z = V_Z I_Z \leq P_{Z(max)}$.

Example: V_{in} = 15 V, V_Z = 10 V, I_L = 20 mA, $I_{Z(min)}$ = 5 mA → R_S = 5/25 mA = 200 Ω.

2. Special Purpose Diodes

Diode	Construction / Principle	Key features / Uses
Varactor (Varicap)	Reverse biased diode ki **transition capacitance** voltage se vary hoti hai. $C_T = K/(V_0+V_R)^n$, n = 1/2 (abrupt), 1/3 (graded).	Voltage-controlled capacitor: TV/FM tuning, VCO, AFC, parametric amplifier.
Point-contact	N-type Ge/Si chip par fine **tungsten/gold wire (cat's whisker)** press; current pulse se chhota P-region banta hai.	Junction area bahut chhota → very low capacitance → **high frequency / microwave detection**, mixers.
Tunnel (Esaki)	Very heavy doping (~1 in 10^3), depletion layer ~10 nm → electrons barrier ke **through tunnel** karte hain. V-I mein I_P (peak) se I_V (valley) tak **negative resistance region**.	Very fast switching, microwave oscillators/amplifiers. Reverse mein bhi conduct karta hai.
Schottky barrier	**Metal–semiconductor** (metal + N-type) junction; current sirf **majority carriers** se, minority charge storage nahi.	Low V_γ (0.2–0.3 V), negligible reverse recovery time → high-speed switching, SMPS, clamping, RF.
LED	Forward bias mein electron-hole **recombination** par energy **photon** ke roop mein nikalti hai. Direct band gap materials: GaAs (IR), GaAsP (red/yellow), GaP (green), GaN (blue). $\lambda(nm) \approx 1240/E_g(eV)$.	V_F ≈ 1.5–3.5 V; series resistor zaroori. Indicators, displays (7-segment), optocouplers. Low power, long life.

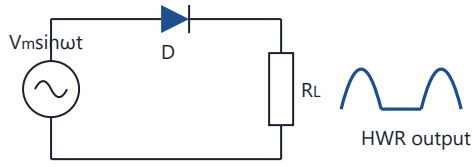


Tunnel diode V-I (negative resistance between I_P and I_V)

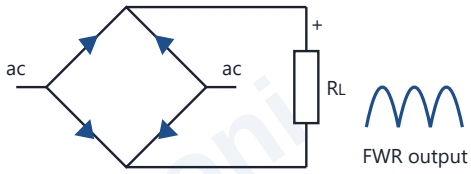
3. Power Rating of Diodes

Diode mein dissipated power $P_D = V_D \times I_D$. Maximum power rating $P_{D(max)}$ wo limit hai jiske upar junction temperature safe limit (Si ≈ 150–200 °C) se zyada ho jaata hai. Ambient temperature badhne par rating ko **derate** karte hain: $P_D(T) = P_D(25\,°C) - (\text{derating factor})(T - 25)$. Other ratings: $I_{F(max)}$, PIV/PRV, surge current. Example: Zener 1 W, 5 V → $I_{Z(max)}$ = 200 mA.

4. Rectifiers (Half Wave & Full Wave)



Half wave rectifier



Bridge full wave rectifier (4 diodes)

HWR: sirf positive half cycle mein diode conduct karta hai. **Centre-tap FWR:** 2 diodes, centre-tapped transformer; har half cycle mein ek diode conduct. **Bridge FWR:** 4 diodes, har half mein 2 diodes series mein conduct; centre tap nahi chahiye.

Parameter	Half Wave	FW Centre-tap	FW Bridge
I_{dc} / V_{dc}	I_m/π, V_m/π	$2I_m/\pi$, $2V_m/\pi$	$2I_m/\pi$, $2V_m/\pi$
I_{rms}	$I_m/2$	$I_m/\sqrt{2}$	$I_m/\sqrt{2}$
Efficiency η (max)	40.6 %	81.2 %	81.2 %
Ripple factor γ	1.21	0.482	0.482
PIV of diode	V_m	$2V_m$	V_m
Ripple frequency	f	2f	2f
TUF	0.287	0.693	0.812

FORMULA

$$\eta = \frac{P_{dc}}{P_{ac}} = \frac{I_{dc}^2 R_L}{I_{rms}^2 (r_f + R_L)};\quad \text{HWR: } \eta = \frac{0.406}{1 + r_f/R_L};\quad \gamma = \frac{I_{ac(rms)}}{I_{dc}} = \sqrt{\left(\left(I_{rms}/I_{dc}\right)^2 - 1\right)}$$

Form factor = rms/avg (HWR 1.57, FWR 1.11); Peak factor = peak/rms (HWR 2, FWR √2).

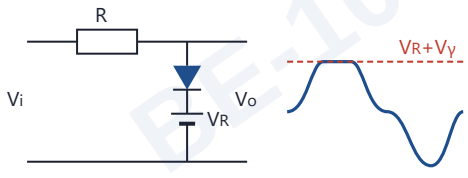
Rectifiers with filter

Filter	Working	Ripple factor
Shunt capacitor (C)	C peak tak charge, diode OFF hone par R_L mein discharge; light load ke liye best	HWR: $1/(2\sqrt{3}\, fCR_L)$; FWR: $1/(4\sqrt{3}\, fCR_L)$
Series inductor (L)	L current change ko oppose karta hai; heavy load ke liye	FWR: $R_L/(3\sqrt{2}\,\omega L)$
LC (choke input) / π (CLC)	Dono ka combination; ripple load par lagbhag independent (LC)	LC: $\sqrt{2}/(12\omega^2 LC) = 1/(6\sqrt{2}\,\omega^2 LC)$

Capacitor filter ke saath $V_{dc} \approx V_m - I_{dc}/(4fC)$ (FWR), aur no-load par $V_{dc} = V_m$.

5. Clippers (Limiters)

Clipper input waveform ke ek hissa (reference level ke upar ya neeche) ko **kaat deta hai** bina baaki shape badle. Types: **series** (diode load ke series mein) & **shunt/parallel** (diode load ke across); **positive**, **negative**, **biased** (battery V_R ke saath) aur **combination (double-ended)** clipper (do diode + do battery, dono side clip). Shunt positive clipper: $V_{in} > V_R + V_\gamma$ → diode ON → $V_o = V_R + V_\gamma$; else $V_o = V_{in}$.



Biased shunt positive clipper & output

6. Clampers (DC Restorers)

Clamper waveform ki shape same rakhkar usko **DC level shift** karta hai (C + diode + R). Capacitor peak tak charge hota hai aur fir battery ki tarah behave karta hai. Condition: $\tau = RC \gg T/2$.

- **Positive clamper:** waveform upar shift; negative peak 0 par clamp → output 0 se $2V_m$.
- **Negative clamper:** waveform neeche shift → 0 se $-2V_m$.
- **Biased clamper:** battery se clamping level V_R par.

7. Voltage Multipliers

Diodes + capacitors se transformer ke peak V_m ka multiple DC output (high voltage, low current). **Half-wave doubler:** C_1 V_m tak charge (D_1), next half mein source + C_1 series se C_2 = **$2V_m$** (D_2 ka PIV = $2V_m$). **Full-wave doubler:** C_1, C_2 alternate halves mein V_m charge, output series = $2V_m$. **Tripler/Quadrupler:** aur stages jodkar $3V_m$, $4V_m$ (n stages → nV_m). Uses: CRT anode supply, X-ray, photocopier.

EXAM TIP Clipper = shape change (part cut). Clamper = shape same, sirf DC level shift. Clamper mein capacitor zaroori, clipper mein nahi.

UNIT III BJT & FET

1. BJT Basic Concept

Bipolar Junction Transistor = 3 layer (NPN ya PNP), 2 junctions, 3 terminals: **Emitter** (heavily doped), **Base** (very thin, lightly doped), **Collector** (moderately doped, largest). "Bipolar" kyunki electrons aur holes dono conduction mein. Active mode: E-B junction forward, C-B junction reverse. Emitter se injected carriers ka ~98–99 % base cross karke collector tak pahunchta hai.

FORMULA

$$I_E = I_B + I_C$$

$$\alpha = I_C/I_E,\ \ \beta = I_C/I_B,\ \ \beta = \frac{\alpha}{1-\alpha},\ \ \alpha = \frac{\beta}{1+\beta},\ \gamma = 1+\beta$$

$$I_C = \alpha I_E + I_{CBO} = \beta I_B + (1+\beta) I_{CBO}$$

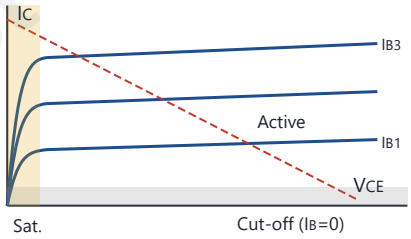
$I_{CEO} = (1+\beta)I_{CBO}$.

2. Input & Output Characteristics of All Configurations

Config.	Input char.	Output char.	Z_{in} / Z_{out}	Gain	Use
CB	I_E vs V_{EB} (V_{CB} const) — diode jaisa; V_{CB} badhe to curve left shift (Early)	I_C vs V_{CB} (I_E const); $I_C \approx I_E$, nearly flat	Very low (~50 Ω) / very high	$A_i = \alpha < 1$, A_v high	HF amplifier
CE	I_B vs V_{BE} (V_{CE} const); V_{CE} badhe to curve right shift	I_C vs V_{CE} (I_B const); slight slope (Early effect)	Medium (~1 kΩ) / medium-high	$A_i = \beta$, A_v high, 180° phase shift	Most common amplifier
CC (emitter follower)	I_B vs V_{CB} (V_{CE} const)	I_E vs V_{CE} (I_B const)	Very high / very low	$A_i = 1+\beta$, $A_v \approx 1$	Buffer, impedance matching

3. Regions of Operation

Region	E-B jn	C-B jn	Use
Active	FB	RB	Amplifier
Saturation	FB	FB	Switch ON
Cut-off	RB	RB	Switch OFF
Reverse active	RB	FB	Rarely used



CE output characteristics with DC load line

4. Biasing & Thermal Stabilization

Biasing = DC voltages lagana taaki Q-point active region ke beech mein fix rahe (faithful amplification). Q-point temperature ke saath shift hota hai kyunki **I_{CBO}** (har 10 °C par double), **V_{BE}** (−2.5 mV/°C) aur **β** badalte hain. **Stabilization** = in changes ke bawajood I_C constant rakhna.

STABILITY FACTOR

$$S = \frac{\partial I_C}{\partial I_{CBO}} = \frac{1+\beta}{1-\beta\,(\partial I_B/\partial I_C)}$$

Ideal S = 1; S jitna kam, stability utni achhi. Also $S' = \partial I_C/\partial V_{BE}$, $S'' = \partial I_C/\partial \beta$.

Bias circuit	Key equations	Stability factor S
Fixed bias (base resistor)	$I_B = (V_{CC} - V_{BE})/R_B$	$1 + \beta$ (poor)
Collector-to-base (feedback) bias	$I_B = (V_{CC} - V_{BE})/(R_B + \beta R_C)$	$(1+\beta)/[1 + \beta R_C/(R_C+R_B)]$ (better)
Voltage divider (self) bias	$V_{Th} = V_{CC}R_2/(R_1+R_2)$, $R_{Th} = R_1 \| R_2$; $I_B = (V_{Th} - V_{BE})/(R_{Th} + (1+\beta)R_E)$	$(1+\beta)(1 + R_{Th}/R_E)/(1+\beta+R_{Th}/R_E) \approx 1 + R_{Th}/R_E$ (best)

Voltage divider bias mein R_E **negative feedback** deta hai: $I_C\uparrow \rightarrow V_E\uparrow \rightarrow V_{BE}\downarrow \rightarrow I_B\downarrow \rightarrow I_C\downarrow$. Isliye ye β-independent aur most stable hai.

5. AC & DC Load Line and Analysis

DC load line: $V_{CE} = V_{CC} - I_C(R_C+R_E)$. End points: $(V_{CC}, 0)$ aur $(0, V_{CC}/(R_C+R_E))$. Slope = $-1/R_{DC}$.

AC load line: AC mein capacitors short → $R_{ac} = R_C \| R_L$ (R_E bypassed). Slope = $-1/R_{ac}$ (zyada steep), Q-point se pass hoti hai.

FORMULA AC load line end points:

$$I_{C(sat)} = I_{CQ} + V_{CEQ}/R_{ac}$$

$$V_{CE(cutoff)} = V_{CEQ} + I_{CQ}R_{ac}$$

Maximum symmetrical swing ke liye Q-point AC load line ke **centre** par: $V_{CEQ} = I_{CQ}R_{ac}$.

6. Compensation Techniques & Thermal Runaway

- **Thermal runaway:** $I_C\uparrow \rightarrow$ power dissipation $P_C = V_{CE}I_C\uparrow \rightarrow$ junction temp $T_j\uparrow \rightarrow I_{CBO}\uparrow \rightarrow I_C$ aur ↑ ... ye cumulative cycle transistor ko **jala** sakta hai.
- Avoid karne ki condition: $\partial P_C/\partial T_j < 1/\theta$ (θ = thermal resistance, °C/W). Agar $\mathbf{V_{CE} < V_{CC}/2}$ ho (R_E wale circuit mein) to runaway nahi hota. Heat sink use karke θ kam karte hain.
- **Compensation techniques** (stabilization se alag — temperature-sensitive devices se opposite effect): (i) **Diode compensation** for V_{BE}: emitter/base circuit mein diode jiska V_D bhi −2.5 mV/°C se change hota hai → V_{BE} change cancel; (ii) **Diode compensation for I_{CO}**: reverse biased diode ka leakage I_0 base current se subtract; (iii) **Thermistor** (negative TC) R_2 ki jagah ya R_E ke saath; (iv) **Sensistor** (positive TC resistor).

7. Ebers-Moll Equations & Early Effect

EBERS-MOLL MODEL (NPN)

$$I_E = I_{ES}(e^{V_{BE}/V_T} - 1) - \alpha_R I_{CS}(e^{V_{BC}/V_T} - 1)$$

$$I_C = \alpha_F I_{ES}(e^{V_{BE}/V_T} - 1) - I_{CS}(e^{V_{BC}/V_T} - 1)$$

Reciprocity: $\alpha_F I_{ES} = \alpha_R I_{CS}$ (= I_S). Model = do back-to-back diodes + do dependent current sources ($\alpha_F I_F$, $\alpha_R I_R$). Ye har region (active, saturation, cut-off) mein valid hai.

Early effect (base-width modulation): V_{CB} (ya V_{CE}) badhne par C-B depletion region chauda hota hai aur **effective base width W_B ghat** jaati hai. Result: (i) base recombination kam → α, β badhte hain → I_C V_{CE} ke saath slightly badhta hai (output curves sloped); (ii) input curves shift; (iii) extreme case mein **punch-through**. Output curves ko peeche extend karne par wo **$-V_A$** (Early voltage, ~50–200 V) par milte hain. Output resistance $r_o = V_A/I_C$.

8. BJT as Amplifier & BJT as Switch

Amplifier (CE)

Q-point active region ke beech mein; small AC input base par → I_b change → $I_c = \beta I_b$ → R_C par bada voltage change, 180° phase shift.

$g_m = I_C/V_T$, $r_\pi = \beta/g_m$, $A_v = -g_m(R_C \| R_L) = -\beta(R_C \| R_L)/r_\pi$

Example: I_C = 1 mA → g_m = 38.5 mA/V; R_C = 4 kΩ → $|A_v| \approx 154$.

Switch

OFF: cut-off, $V_{in} = 0 \rightarrow I_C \approx 0$, $V_{out} = V_{CC}$ (open switch). **ON:** saturation, $V_{CE(sat)} \approx 0.2$ V, $I_{C(sat)} = (V_{CC} - V_{CE(sat)})/R_C$ (closed switch). Saturation ke liye $I_B > I_{C(sat)}/\beta$. Ye NOT gate (inverter) jaisa kaam karta hai; relays, LEDs, digital logic drive karta hai.

9. FET: Basic Concept

Field Effect Transistor = **unipolar** (sirf majority carriers), **voltage-controlled** device; gate voltage se channel ka width/conductivity control hota hai. Terminals: Source, Drain, Gate. Very high input impedance (JFET ~10^9 Ω, MOSFET ~10^{12}–10^{15} Ω), low noise, thermally stable (no thermal runaway), chhota size.

JFET

N-channel bar ke dono side P^+ gate regions. Gate-source junction **reverse biased**; V_{GS} zyada negative → depletion chauda → channel patla → I_D kam. $V_{GS} = V_P$ (pinch-off / $V_{GS(off)}$) par $I_D = 0$.

- **Drain char.** (I_D vs V_{DS}): ohmic region → pinch-off (saturation, I_D const) → breakdown.
- **Transfer char.** (I_D vs V_{GS}): parabolic (Shockley).

MOSFET

Gate SiO_2 se insulated (IGFET). **Depletion (D-MOSFET):** channel pehle se bana; $-V_{GS}$ (depletion) aur $+V_{GS}$ (enhancement) dono mode. **Enhancement (E-MOSFET):** koi channel nahi; $V_{GS} > V_T$ (threshold) par inversion layer channel banta hai; normally OFF.

Applications (JFET/MOSFET): buffer/ high-Z input amplifier, analog switch, chopper, VVR (voltage-variable resistor), RF amplifier, CMOS digital ICs, power switching (SMPS, motor drives), memory.

FET FORMULAS JFET / D-MOSFET: $I_D = I_{DSS}(1 - V_{GS}/V_P)^2$; $g_m = g_{m0}(1 - V_{GS}/V_P)$, $g_{m0} = 2I_{DSS}/|V_P|$ | E-MOSFET: $I_D = k(V_{GS} - V_T)^2$ | CS amplifier $A_v = -g_m R_D$.

Example: I_{DSS} = 10 mA, $V_P = -4$ V, $V_{GS} = -2$ V → $I_D = 10(1-0.5)^2 = 2.5$ mA.

BJT vs FET: BJT bipolar, current-controlled, low–medium Z_{in}, thermal runaway possible, higher gain-bandwidth; FET unipolar, voltage-controlled, very high Z_{in}, thermally stable, less noise, smaller & easy IC fabrication.

UNIT IV Operational Amplifier

1. Op-Amp Basics

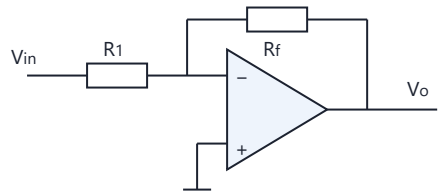
Op-amp = high-gain, direct-coupled **differential amplifier** IC (e.g. IC 741) jisme do inputs: inverting (−) aur non-inverting (+), ek output. Internal stages: differential amp → intermediate gain stage → level shifter → output (push-pull) stage. $V_o = A_{OL}(V_2 - V_1)$.

Virtual short: negative feedback ke saath $V_+ \approx V_-$ aur input current ≈ 0.

Parameter	Ideal	IC 741
Open loop gain A_{OL}	∞	2×10^5
R_{in}	∞	2 MΩ
R_{out}	0	75 Ω
Bandwidth	∞	1 MHz (GBW)
CMRR	∞	90 dB
Slew rate	∞	0.5 V/µs

CMRR = A_d/A_{cm} (dB = 20 log); Slew rate = $(dV_o/dt)_{max}$; Input offset voltage, input bias current = $(I_{B1}+I_{B2})/2$.

2. Inverting & Non-Inverting Modes



Inverting amplifier: $A_v = -R_f/R_1$

OP-AMP GAINS

$$\text{Inverting: } A_v = \frac{V_o}{V_{in}} = -\frac{R_f}{R_1}, \quad R_{in} = R_1$$

$$\text{Non-inverting: } A_v = 1 + \frac{R_f}{R_1}, \quad R_{in} \text{ very high}$$

Voltage follower (R_f=0, R_1=∞): A_v = 1 (buffer). Summer: $V_o = -R_f(V_1/R_1 + V_2/R_2 + \ldots)$.

Derivation (inverting): $V_- \approx 0$ (virtual ground). Current $I = V_{in}/R_1 = (0 - V_o)/R_f \rightarrow V_o = -(R_f/R_1)V_{in}$. **Non-inverting:** $V_- = V_{in} = V_o R_1/(R_1+R_f) \rightarrow V_o = (1+R_f/R_1)V_{in}$. Example: R_1 = 10 kΩ, R_f = 100 kΩ → inverting gain −10, non-inverting gain 11.

3. Comparators & Zero Crossing Detector

Comparator: op-amp open loop (no feedback) mein; ek input par signal, doosre par reference V_{ref}. $V_{in} > V_{ref}$ (non-inverting comparator) → $V_o = +V_{sat}$; $V_{in} < V_{ref} \rightarrow -V_{sat}$ ($V_{sat} \approx V_{CC}$ − 1 to 2 V). Uses: level detector, ADC, PWM. **Zero crossing detector (ZCD):** comparator with $V_{ref} = 0$; har baar input zero cross kare tab output state badalta hai → **sine wave to square wave converter**. Drawback: noisy input par zero ke paas multiple false triggering → solution = Schmitt trigger.

4. Schmitt Trigger (Regenerative Comparator)

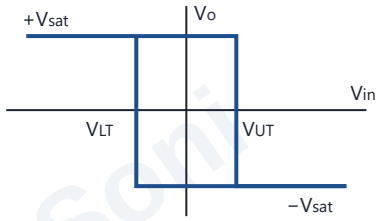
Comparator + **positive feedback** (R_1, R_2 divider se output ka hissa + input par). Do threshold levels → **hysteresis**; noise immunity. Inverting Schmitt mein V_{in} −ve terminal par.

FORMULA

$$V_{UT} = +V_{sat}\frac{R_2}{R_1+R_2},\quad V_{LT} = -V_{sat}\frac{R_2}{R_1+R_2}$$

$$V_H = V_{UT} - V_{LT} = 2V_{sat}R_2/(R_1+R_2)$$

(R_2 = feedback divider ka ground-side resistor.)



Hysteresis loop (inverting Schmitt trigger)

5. Instrumentation Amplifier

Low-level transducer signals (strain gauge, thermocouple) ko amplify karne ke liye **high-gain, high input impedance, very high CMRR** differential amplifier. Standard **3 op-amp** circuit: do non-inverting **buffer** stages (A_1, A_2) jinke beech gain resistor R_G, fir ek **difference amplifier** (A_3). Gain sirf ek resistor R_G badal kar adjust hota hai.

FORMULA

$$V_o = \frac{R_3}{R_2}\left(1+\frac{2R_1}{R_G}\right)(V_2 - V_1)$$

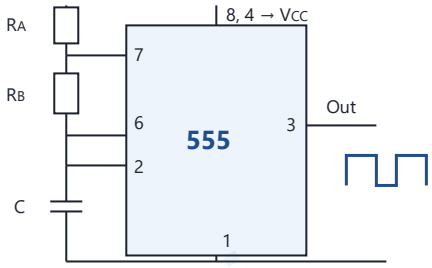
R_1 = buffer feedback resistors; R_2, R_3 = difference-amp resistors. Features: low output impedance, low drift, gain accurately settable. Example IC: AD620, INA128.

6. 555 Timer & Its Applications

IC 555 (8-pin) = precision timing IC. Internal blocks: **3 equal 5 kΩ resistors** divider (refs **2/3 V_{CC}** & **1/3 V_{CC}**), 2 comparators (upper/threshold, lower/trigger), **SR flip-flop**, discharge transistor, output buffer.

Pins: 1 GND, 2 Trigger, 3 Output, 4 Reset, 5 Control voltage, 6 Threshold, 7 Discharge, 8 V_{CC} (4.5–18 V).

Applications: astable/monostable/bistable MV, PWM, PPM, missing pulse detector, frequency divider, LED flasher, timers, tone generator, ramp generator.



555 astable connection

7. Multivibrators

Type	Stable states	Working (555)	Formula
Astable (free-running)	0 (dono quasi-stable)	C, R_A+R_B se 1/3 → 2/3 V_{CC} charge (output HIGH), R_B se pin 7 dwara discharge (LOW). Trigger nahi chahiye; square wave generator/clock.	$t_H = 0.693(R_A+R_B)C$, $t_L = 0.693R_BC$ **$f = 1.44/[(R_A+2R_B)C]$** $D = (R_A+R_B)/(R_A+2R_B)$
Monostable (one-shot)	1	Trigger pin 2 par negative pulse (< 1/3 V_{CC}) → output HIGH; C 0 se 2/3 V_{CC} charge hone tak, fir wapas LOW (stable). Timer, pulse stretcher.	**$T = 1.1\,RC$**
Bistable (flip-flop)	2	Trigger (pin 2) par SET (HIGH), reset/threshold par RESET (LOW); state tab tak rehta hai jab tak external trigger na aaye. Memory element.	No timing RC

Example: R_A = 1 kΩ, R_B = 10 kΩ, C = 0.1 µF → f = 1.44/(21 k × 0.1 µ) ≈ **686 Hz**, D = 11/21 ≈ 52.4 %. Monostable R = 10 kΩ, C = 10 µF → T = 0.11 s. Op-amp astable (square wave): T = 2RC ln[(1+β)/(1−β)], β = $R_2/(R_1+R_2)$.

COMMON MISTAKE 555 astable mein duty cycle hamesha > 50 % hota hai ($t_H > t_L$). 50 % ke liye R_B ke across diode lagate hain ya $R_B \gg R_A$ lete hain.

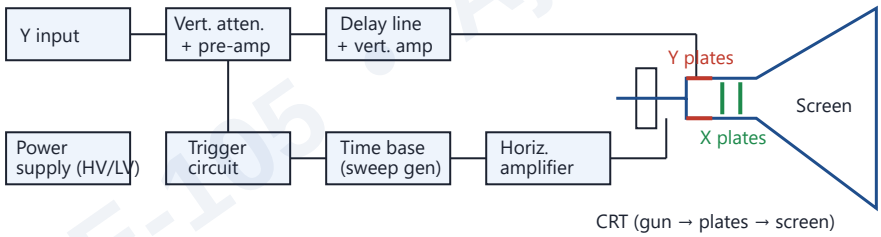
UNIT V CRO & Transducers

1. Cathode Ray Oscilloscope (CRO) — Construction

CRO ek instrument hai jo voltage waveform ko time ke saath screen par **display** karta hai; amplitude, time period, frequency, phase measure karte hain. Dil hai **CRT (cathode ray tube)** — evacuated glass tube with:

- **Electron gun:** heater + cathode (thermionic emission), **control grid** (intensity/brightness — negative potential), pre-accelerating, focusing & accelerating anodes.
- **Deflection system:** vertical (Y) plates aur horizontal (X) plates.
- **Fluorescent screen:** phosphor coating (P31 green etc.); electron strike → light. **Aquadag** (graphite) coating secondary electrons collect karti hai.
- **Glass envelope:** high vacuum, taaki electrons gas molecules se na takrayen.

2. Block Diagram of CRO



CRO block diagram: vertical section, horizontal (time-base) section, trigger, CRT, power supply

Delay line signal ko ~0.25 µs delay karti hai taaki sweep start hone ke baad signal Y plates par pahunche (leading edge dikh jaye). **Time base** sawtooth (ramp) voltage X plates par deta hai → spot left se right uniformly move karta hai, fir quick flyback (blanked). **Trigger** sweep ko signal ke same point se start karata hai → stable display.

3. Deflection & Focusing System

Electrostatic deflection

Plates ke beech field E = V_d/d electron ko parabolic path mein mod deta hai; plates ke baad seedha screen tak.

FORMULA

$$D = \frac{L\,l\,V_d}{2\,d\,V_a}, \quad S = \frac{D}{V_d} = \frac{L\,l}{2\,d\,V_a}\ \text{m/V}$$

L = plates centre se screen distance, l = plate length, d = plate separation, V_a = accelerating voltage. Deflection factor G = 1/S (V/m). Electron velocity v = $\sqrt{(2eV_a/m)}$.

Focusing

Electrostatic focusing: focusing anode aur accelerating anodes ke beech ka non-uniform field **electrostatic lens** banata hai jo divergent beam ko screen par ek sharp spot par converge karta hai (focus control = focusing anode potential adjust). **Magnetic focusing** (coil) TV CRTs mein. Magnetic deflection (coils) bhi possible: D ∝ $1/\sqrt{V_a}$, large-angle (TV), jabki CRO mein electrostatic (high frequency response).

Example: L = 20 cm, l = 2 cm, d = 0.5 cm, V_a = 1000 V → S = (0.2×0.02)/(2×0.005×1000) = 0.4 mm/V.

4. Lissajous Figures

Jab Y aur X dono plates par sinusoidal signals (time base OFF) diye jaate hain to screen par bana pattern = **Lissajous figure**. Use: unknown **frequency** aur **phase difference** measure karna.

FORMULA

$$\frac{f_y}{f_x} = \frac{\text{No. of horizontal tangencies}}{\text{No. of vertical tangencies}}$$

$$\text{Phase: } \sin\varphi = Y_1/Y_2$$

Y_1 = ellipse ka y-axis intercept, Y_2 = max vertical deflection.

Phase φ ($f_y = f_x$)	Pattern
0° / 180°	Straight line (right / left tilt)
0°–90° / 90°–180°	Tilted ellipse (right / left)
90° (equal amplitudes)	Circle
f_y:f_x = 2:1	Figure-of-eight type (2 horizontal, 1 vertical tangency)

5. Dual Trace & Dual Beam CRO

Dual Trace CRO	Dual Beam CRO
Ek electron gun/beam; electronic switch dono channels (A, B) ko ek hi vertical amp par switch karta hai.	**Do** alag electron beams (do guns ya split beam), alag vertical plates/amps.
Modes: **Alternate** (ek sweep mein A, next mein B — high freq) & **Chopped** (sweep ke dauraan ~100–500 kHz par fast switching — low freq). Sasta, common; single-shot events ke liye nahi.	Dono signals truly simultaneously display; fast single-shot transients capture. Mehenga, bada CRT; same ya alag time base.

6. Sampling & Storage Oscilloscopes

Sampling oscilloscope

Bahut **high frequency** (GHz) repetitive signals ke liye. Har cycle mein signal ka ek **sample** leta hai, har agle cycle mein sample point thoda aage shift (sequential sampling). Samples jodkar slow "reconstructed" waveform display hota hai. Condition: signal **repetitive** ho.

Storage oscilloscope

Waveform ko screen par lambe time tak **store** karta hai — slow/one-time transients ke liye. **Analog storage:** special CRT — **bistable** storage aur **variable persistence** (storage mesh + flood gun). **Digital storage oscilloscope (DSO):** signal → ADC → digital memory → DAC → display; waveform save, analyse, PC transfer.

7. Classification of Transducers

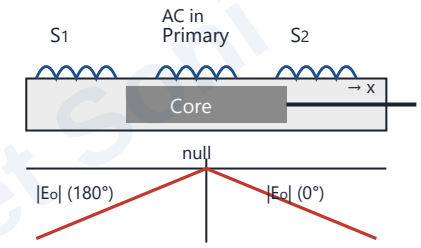
Transducer = device jo ek form ki energy/physical quantity (pressure, displacement, temp.) ko doosri form (mostly electrical) mein convert kare.

Basis	Types & Examples
Power	**Active** (self-generating, external power nahi): thermocouple, piezoelectric, photovoltaic cell. **Passive** (external excitation chahiye): strain gauge, LVDT, thermistor, RTD, capacitive.
Principle	Resistive (strain gauge, POT), Inductive (LVDT), Capacitive, Piezoelectric, Photoelectric, Thermoelectric
Prim./Sec.	Primary: physical qty → intermediate (Bourdon tube, diaphragm). Secondary: intermediate → electrical (LVDT).
Output / Direction	**Analog** (LVDT, thermocouple) vs **Digital** (shaft encoder); **Transducer** (non-elec → elec) vs **Inverse** (elec → non-elec: loudspeaker)

8. LVDT (Linear Variable Differential Transformer)

Passive **inductive** transducer jo **linear displacement** ko voltage mein badalta hai. Ek **primary** (centre) + do identical **secondaries** S_1, S_2 (**series opposition**); beech mein movable soft-iron core (shaft se juda). Primary par AC (1–10 kHz).

- **Null position** (core centre): $E_{S1} = E_{S2} \rightarrow E_o = 0$.
- Core S_1 ki taraf → $E_{S1} > E_{S2}$, output in phase; S_2 ki taraf → output 180° out of phase. Magnitude ∝ displacement (linear range), **phase se direction**.



LVDT construction and output vs displacement

LVDT

$$E_o = E_{S1} - E_{S2} \propto x$$

Adv: frictionless, high resolution & output, linear, rugged, low hysteresis. **Disadv:** stray magnetic field se affect, AC excitation + demodulator chahiye. **Uses:** displacement, force, pressure.

9. Strain Gauge

Passive **resistive** transducer. Fine wire/foil grid ko object par chipkate hain; strain se length badhti & area ghatta hai (piezoresistive effect se ρ bhi badalta hai) → resistance change. Δ R ko **Wheatstone bridge** se measure karte hain. Types: unbonded, bonded (wire/foil), semiconductor (G ≈ 100–200, temperature sensitive). Temperature compensation ke liye **dummy gauge** (unstrained) bridge ki adjacent arm mein.

GAUGE FACTOR

$$G = \frac{\Delta R/R}{\Delta L/L} = \frac{\Delta R/R}{\varepsilon} = 1 + 2\nu + \frac{\Delta\rho/\rho}{\varepsilon}$$

ν = Poisson's ratio; metal gauge G ≈ 2. **Example:** R = 120 Ω, G = 2, ε = 500×10^{-6} → ΔR = GRε = $2\times120\times500\times10^{-6}$ = 0.12 Ω.